

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0033

Weslaco 230 kV
October 20, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0033. The Interconnection Customer proposes a 158.3 MW battery energy storage facility interconnecting at the Weslaco 230 kV substation in the Illinois area. The request was received on May 06, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 29 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,396,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Darlington Rd / Wayne Trace 100 kV L: 118.2% loading (111 MVA) following the outage of the Richland / Dekalb 500 kV Line; DFAX = 6.0%.

For the loss of Richland / Dekalb 500 kV, the Muskogee / Darlington Road 100 kV L loads to 113.5% of its 179 MVA rating. Project DFAX: 6.8%.

For the loss of Dekalb / Richland 500 kV, the Waynedale / State St 230 kV loads to 105.3% of its 572 MVA rating. Project DFAX: 17.9%.

The Dairymans / Pueblo Xtra 230 kV is overloaded to 103.1% of its 725 MVA emergency rating for the loss of the Pueblo Xtra / Dairymans 230 kV Line. The project contributes 9.5% DFAX (15.0 MW) to this overload.

For the loss of Flood St / Cook 500 kV, the Flood Street / University 230 kV loads to 102.0% of its 909 MVA rating. Project DFAX: 17.0%.

For the loss of Waynedale / Montpelier Electric Generating Sta 230 kV, the Monfort / Waynedale 230 kV line loads to 101.9% of its 700 MVA rating. Project DFAX: 12.0%.

Dekalb / Slocum Rd 230 kV: 101.2% loading (508 MVA) following the outage of the Dekalb / Richland 500 kV; DFAX = 26.0%.

Mcneil Substation / Slocum Rd 230 kV: 101.1% loading (536 MVA) following the outage of the Richland / Dekalb 500 kV line; DFAX = 26.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Darlington Rd / Wayne Trace 100 kV L	Rebuild line with higher-capacity conductor	\$2,870,000	15% (\$422,000)
Muskogee / Darlington Road 100 kV L	Rebuild line with higher-capacity conductor	\$22,583,000	15% (\$3,323,000)
State St / Waynedale 230 kV	Reconductor line	\$6,657,000	15% (\$984,000)
Dairymans / Pueblo Xtra 230 kV	Reconductor line	\$24,289,000	36% (\$8,751,000)
Flood St / University 230 kV	Reconductor line	\$9,191,000	35% (\$3,183,000)
Waynedale / Monfort 230 kV L	Reconductor line	\$12,988,000	14% (\$1,836,000)
Dekalb / Slocum Rd 230 kV Line	Reconductor line	\$13,881,000	15% (\$2,108,000)
Slocum Rd / Mcneil 230 kV	Reconductor line	\$5,197,000	15% (\$789,000)
	Total allocated to project		\$21,396,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.